1 Static Yao-Kivelson Model

1.1 Kitaev Model and Yao-Kivelson Model

Kitaev introduced a way to map the spin liquid model

$$H = \sum_{\substack{\text{x-bonds} \\ \langle jk \rangle}} J^x_{jk} \sigma^x_j \sigma^k_x + \sum_{\substack{\text{y-bonds} \\ \langle jk \rangle}} J^y_{jk} \sigma^y_j \sigma^k_y + \sum_{\substack{\text{z-bonds} \\ \langle jk \rangle}} J^z_{jk} \sigma^z_j \sigma^k_z, \tag{1.1}$$

now known as the Kitaev Model, to a free Majorana-Fermion hopping model, and therefore managed to solve the model exactly [1]. Classically, 'the ' Kitaev model is defined on a honeycomb lattice, where each coupling type $\alpha = x, y, z$ corresponds to one of the three bond orientations. However, his method directly generalizes to any lattice geometry [2], provided that each site has exactly one bond of each type $\alpha = x, y, z$.

One alternative to the honeycomb lattice is the star/triangle-honeycomb lattice. Instead of elementary plaquettes of size 6, it features two different types of plaquettes of size 3 and 12, respectively. When defined on this geometry, model 1.1 is known as the Yao-Kivelson Model [3]. Due to the odd length of the triangular plaquettes, this model features a groundstate that spontaneously breaks Time-Reversal Symmetry, and therefore describes a Chiral Spin Liquid.

1.2 Kitaev's Spin-Majorana Mapping

Kitaev introduced a representation of a spin-$\frac{1}{2}$-degree of freedom (called 'spin' in the following) in terms of 4 fermionic Majorana modes. Consider the 4-dimensional Fockspace $\mathcal{H}$ of two abstract fermions a_1, a_2 satisfying the canonical anti-commutation relations

$$\left\{a_k, a_l^\dagger\right\} = \delta_{kl} \quad \text{and} \quad \left\{a_k, a_l\right\} = 0 \quad \text{for} \quad k, l = 1, 2. \tag{1.2}$$

They give rise to the 4 operators

$$c_{2k-1} \coloneqq c_k + c_k^\dagger \quad \text{and} \quad c_{2k} \coloneqq \frac{1}{i}\Big(a_k - a_k^\dagger\Big), \qquad \text{for} \qquad k = 1, 2, \tag{1.3}$$

which are commonly called Majorana modes. They satisfy

$$c_i^\dagger = c_i \quad \text{and} \quad \big\{c_i, c_j\big\} = 2\delta_{ij} \quad \text{for} \quad i, j = 1, \ldots, 4, \tag{1.4}$$

which qualifies them to be called "real fermions", in contrast to the usual complex ones.

These will be used to represent the spin algebra (see sec. 2.1), i.e. construct a set $\{\sigma^\alpha\}_{\alpha=x,y,z}$ of 2-dim. hermitian operators obeying

$$\sigma^\alpha \sigma^\beta = \delta^{\alpha\beta} + i\epsilon^{\alpha\beta\gamma}\sigma^\gamma, \qquad \alpha, \beta, \gamma = x, y, z. \tag{1.5}$$

For that, rename the 4 Majorana modes from $\{c_i\}_{i=1,\ldots,4}$ to $\{b^x, b^y, b^z, d\}$ and define

$$\tilde{\sigma}^\alpha \coloneqq i b^\alpha d, \qquad \alpha = x, y, z. \tag{1.6}$$

Direct calculation shows that this definition suffices eq. 1.5 if and only if $\tilde{\sigma}^x\tilde{\sigma}^y\tilde{\sigma}^z \overset{!}{=} i$ holds, which is equivalent to

$$1 \overset{!}{=} b^x b^y b^z d =: D, \tag{1.7}$$

where we have defined the hermitian operator D. Due to $D^2 = 1$, $\mathcal{H}$ splits into the eigenspaces $\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_{-1}$, where the subscript denotes the respective eigenvalues of D. $\mathcal{H}_1$ is the space where eq. 1.7 is fulfilled, and is therefore called the 'physical' subspace, while $\mathcal{H}$ is called 'extended' Hilbert space. D cannot be the identity on $\mathcal{H}$, as it anti-commutes with each Majorana operator:

$$\{D, c_i\} = 0 \quad \text{for} \quad c_i = b^x, b^y, b^z, d. \tag{1.8}$$

This anti-commutation implies that each Majorana operator c_i maps 1-to-1 between $\mathcal{H}_1$ and $\mathcal{H}_{-1}$, which shows that $\dim(\mathcal{H}_1) = \dim(\mathcal{H}_{-1}) = 2$. Equation 1.8 further implies that D commutes with operators consisting of an even number of Majorana operators. In particular,

$$[D, \tilde{\sigma}^\alpha] = 0 \quad \text{for} \quad \alpha = x, y, z, \tag{1.9}$$

which means the $\tilde{\sigma}^\alpha$ preserve the physical subspace and their restriction to $\mathcal{H}_1$,

$$\sigma^\alpha \coloneqq \tilde{\sigma}^\alpha|_{\mathcal{H}_1}, \tag{1.10}$$

is well-defined and finally yields the desired representation of the spin algebra.

Any Hamiltonian describing a spin can be seen as Pauli matrices acting on $\mathcal{H}_1$. It can be beneficial to replace the σ^α by $\tilde{\sigma}^\alpha$ acting on the extended Hilbert space, i.e. allowing states to have unphysical degrees of freedom. Such a state can always be projected into $\mathcal{H}_1$ using the projector

$$P = \frac{1+D}{2} \tag{1.11}$$

to obtain a physical state. If we manage to find an H-eigenstate $|\psi\rangle$ in the extended Hilbert space $\mathcal{H}$, the projection $P|\psi\rangle \in \mathcal{H}_1$ will either be a non-normalized physical eigenstate of the same energy or the zero-vector. This follows from the fact that D and therefore also P commute with H according to eq. 1.9. Hence, in case the problem admits a solution in the extended Hilbert space, it is also solved in the physical space.

To obtain more intuition about the unphysical degrees of freedom, Kitaev used the concept of gauge transformations. Together with the identity, D is a unitary representation of the gauge group $\mathcal{G} = Z_2$ acting on the extended Hilbert space $\mathcal{H}$. This allows the characterization of the physical states $\mathcal{H}_1$ as

$$|\psi\rangle \text{ is physical } \iff |\psi\rangle \text{ is invariant under } \mathcal{G}. \tag{1.12}$$

The projector from eq. 1.11 can be interpreted as the group-average, which is in general known to project into the invariant subspace. If two states are related to each other by the application of D, they are called gauge-equivalent and they correspond to the same physical state. The converse is in general not true: Not all the states corresponding to the same physical state need to be gauge equivalent.

The mapping discussed in this section generalizes straight-forwardly to many-spin systems, where the operators now act on the product space. The gauge symmetry then becomes a local gauge symmetry at each individual site, implying that each gauge-sector contains 2^N gauge-equivalent states in the 4^N-dim. extended Hilbert space of N spins.

1.3 Application of the Mapping to the Model

Applying the spin-Majorana mapping from eq. 1.6 to a general Kitaev Hamiltonian 1.1 yields

$$H = -i \sum_{\substack{\text{bonds} \\ \langle jk \rangle}} J_{jk} u_{jk}^{\alpha(j,k)} \left(d_j d_k\right) \quad \text{with} \quad u_{j,k}^{\alpha} \coloneqq i\, b_j^{\alpha} b_k^{\alpha} = \left(u_{j,k}^{\alpha}\right)^{\dagger}, \tag{1.13}$$

which is grouped into d-Majoranas living on the lattice sites, and pairs of b^{α}-Majoranas living on the bonds. The value $\alpha(j,k) = x, y, z$ should be determined by the bond $\langle jk \rangle$, but this explicit dependence will be omitted from here on. One has to pick some fixed orientation for each bond, as each bond $\langle jk \rangle$ should uniquely identify the operators $\left(d_j d_k\right)$ and $\left(u_{jk}^{\alpha}\right)$, even though they are each anti-symmetric under exchanging $j \leftrightarrow k$.

We now focus on lattices where each site has at most one bond of each type $\alpha = x, y, z$, and denote this requirement by $(*)$. In this case, the $u_{j,k}^{\alpha}$ commute among each other, which can be seen by picking two arbitrary different bonds:

Case 1: The bonds share no site. Then the Majorana modes constituting the two u-operators are 4 independent anti-commuting Majorana modes, i.e. the u-operators are independent commuting Majorana bilinears.

Else: The bonds share exactly one site, w.l.o.g. labeled by $u_{j,k}^{\alpha}$ and $u_{k,l}^{\beta}$ with $j \neq l$. Then $u_{j,k}^{\alpha} u_{k,l}^{\beta} = i^2\, b_j^{\alpha} b_k^{\alpha} b_k^{\beta} b_l^{\beta}$ consists of 4 independent Majorana modes if and only if $\alpha \neq \beta$, which is guaranteed by $(*)$. The claim follows analogously to case 1.

From this, we can conclude that each $u_{j,k}^{\alpha}$ also commutes with the Hamiltonian 1.13: Apart from terms $u_{j,k}^{\alpha}$, H only consists of terms $d_j d_k$, which are Majorana bilinears independent of all the $u_{j,k}^{\alpha}$. With $\left(u_{jk}^{\alpha}\right)^2 = 1$, this means H decomposes into symmetry-sectors labeled by the eigenvalues ± 1 of all the u_{jk}^{α}. In each symmetry sector, the u-operators can then be replaced by their eigenvalue and it remains to solve a quadratic hopping Hamiltonian, which in general possess an accessible exact solution.

However, for the gauge transformation $D_l = b_l^x b_l^y b_l^z d_l$ from eq. 1.7, one finds

$$D_l^{\dagger} u_{jk}^{\alpha} D_l = u_{jk}^{\alpha} \cdot \begin{cases} -1, & j = l \text{ or } j = k \\ +1, & \text{else} \end{cases}, \tag{1.14}$$

i.e. the u_{jk}^{α}-eigenvalues are not gauge invariant. Indeed, for all sites l, a sector labeled by $\left\{u_{jk}^{\alpha}\right\}$ is gauge equivalent to the sector obtained by flipping the u_{jk}^{α}-labels on all bonds

connected to the site l. By gauge-equivalence of vector spaces we mean every state from the one space can be obtained by applying a gauge transformation to a suitable vector from the other space. In our case, this is guaranteed by the fact that D_l is self-inverse, and hence defines a bijection between the two sectors.

To obtain physical observables, we partition the set of labels $\left\{u_{jk}^{\alpha}\right\}$ into gauge-equivalent classes. One can show (see sec. 2.2) that these can be fully characterized by the values of products

$$W_\gamma = \prod_{\langle jk\rangle\in\gamma} \left(-i\, u_{jk}^{\alpha}\right) = \prod_{\langle jk\rangle\in\gamma} \sigma_j^{\alpha}\sigma_k^{\alpha} \tag{1.15}$$

around a set of independent loops γ through the lattice, which are typically the elementary plaquettes. Additionally, there may be non-local non-contractible loops arising from the topology of the underlying manifold, most prominently the two independent non-contractible loops on the torus for the case of periodic boundary conditions.

By fixing the all the values u_{jk}^{α} in Hamiltonian 1.13, we have reduced the problem to free hopping of Majorana modes d_j. Comparing with the loop operators W_γ, we find that W_γ corresponds to a translation operator for the Majorana mode around the loop γ, and therefore measures the flux

$$W_\gamma = \exp\left(i\phi_\gamma\right) \tag{1.16}$$

enclosed by γ. From $u_{jk}^{\alpha} = \pm 1$ and the definition in eq. 1.15 it is clear that we can define flux in the range $\phi \in \left\{0, \pm\frac{\pi}{2}, \pi\right\}$.

Introducing the anti-unitary time reversal operator

$$\mathcal{T} = \left(\prod_j i\sigma_j^y\right) K \tag{1.17}$$

with complex conjugation in z-basis K and using its property

$$\sigma_j^{\alpha} \xrightarrow{\mathcal{T}} -\sigma_j^{\alpha} \qquad \text{for all } j \text{ and } \alpha, \tag{1.18}$$

we see that $\mathcal{T}$ commutes with all the loop operators and the Hamiltonian

$$[H, \mathcal{T}] = [W_\gamma, \mathcal{T}] = 0. \tag{1.19}$$

However, $\mathcal{T}$ maps the flux to its complex conjugate $\phi_\gamma \xrightarrow{\mathcal{T}} \phi_\gamma^*$, which shows that loop-eigenstates with $\pm\frac{\pi}{2}$-flux form degenerate Kramer's pairs with spontaneously broken

time reversal symmetry. Loops have $\pm\frac{\pi}{2}$-flux if and only if they contain an odd number of sites. To see this, note that as a product of u^{α}_{jk} along a closed loop, the loop operator W_γ if a product of pairwise commuting terms, as neighbouring bonds have different bond types $\alpha \neq \alpha'$ due to $*$. Therefore,

$$W_\gamma^\dagger = (-1)^{|\gamma|}\, W_\gamma \tag{1.20}$$

can be read off the factors of i in eq. 1.15, where $|\gamma|$ means the number of sites in γ. This shows that W_γ is purely real/imaginary on even/odd loops.

2 Additional Notes

2.1 Representations of the Pauli Algebra

If we introduce 3 hermitian operators $\{\tilde{\sigma}^x, \tilde{\sigma}^y, \tilde{\sigma}^z\}$ on a 2-dim. Hilbert space, what are the precise conditions for these to represent a spin unitarily equivalently to the usual Pauli matrices? Indeed, it is enough to check that the relation

$$\tilde{\sigma}^\alpha \tilde{\sigma}^\beta = \delta^{\alpha\beta} + i\epsilon^{\alpha\beta\gamma}\tilde{\sigma}^\gamma, \qquad \alpha, \beta, \gamma = x, y, z. \tag{2.1}$$

holds according to the following lemma.

Lemma 1. *All 2-dim. hermitian representations of the Pauli algebra are unitarily equivalent, i.e. if some hermitian 2-dim. operators $\{\tilde{\sigma}^x, \tilde{\sigma}^y, \tilde{\sigma}^z\}$ suffice equation 2.1, there exists a unitary transformation U such that*

$$\tilde{\sigma}^\alpha = U^\dagger \sigma^\alpha U \quad \textit{for} \quad \alpha = x, y, z, \tag{2.2}$$

where σ^α are the usual Pauli matrices in some fixed basis.

Hence, if a set of matrices $\{\tilde{\sigma}^\alpha\}$ suffices these requirements, we can transform any spin Hamiltonian H to $\tilde{H}$ by replacing all the σ^α by $\tilde{\sigma}^\alpha$. This just amounts to picking a certain basis in the abstract Hilbert space.

Proof. We construct a basis in which the matrix representation of the $\tilde{\sigma}^\alpha$ agrees with the usual Pauli matrices. From eq. 2.1 for $\alpha = \beta$ it follows that all the $\tilde{\sigma}^\alpha$ can only have eigenvalues ± 1. As they mutually anti-commute, none of them can be proportional to the identity, hence each one has the spectrum $\{-1, +1\}$. Let $|\uparrow\rangle$ be an eigenvector of $\tilde{\sigma}^z$ with eigenvalue 1, and construct $|\downarrow\rangle \coloneqq \tilde{\sigma}^x |\uparrow\rangle$. We find

$$\tilde{\sigma}^z |\downarrow\rangle = \tilde{\sigma}^z \tilde{\sigma}^x |\uparrow\rangle = -\tilde{\sigma}^x \tilde{\sigma}^z |\uparrow\rangle = -\tilde{\sigma}^x |\uparrow\rangle = -|\downarrow\rangle \tag{2.3}$$

and

$$\tilde{\sigma}^x \left|\downarrow\right\rangle = \left(\tilde{\sigma}^x\right)^2 \left|\uparrow\right\rangle = \left|\uparrow\right\rangle, \tag{2.4}$$

hence in the basis $\{\left|\uparrow\right\rangle, \left|\downarrow\right\rangle\}$, $\tilde{\sigma}^\alpha$ can be written as matrices

$$\tilde{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \tilde{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{and} \quad \tilde{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \tag{2.5}$$

where the matrix representation for $\tilde{\sigma}^y$ is fixed by the condition $\tilde{\sigma}^z\tilde{\sigma}^x = i\tilde{\sigma}^y$. Thus, the $\tilde{\sigma}^\alpha$ agree with the usual Pauli matrices in a suitable basis, which means they have to be related to each other by a unitary transformation. □

2.2 Characterization of Gauge-Equivalent Sectors

In this section, we will translate the lattice geometry into a graph-theoretical language and prove that the equivalence classes of gauge-equivalent states are always uniquely characterized by the flux through the independent loops. The proof is inspired by a more general result on exactly solvable spin models [4] and by well-known results from graph theory [5].

The lattice defines a graph $\mathcal{G}$ with a vertex for each spin. We add an edge between the vertices j and k for every interaction term $\langle jk\rangle$ in the Hamiltonian. Let n be the number of vertices and m be the number of edges. We may further assume that $\mathcal{G}$ is connected, as multiple connected components can be treated independently. The configurations of bond-variables u^α_{jk} correspond to m-tuples of binary variables

$$\boldsymbol{u} = (u_1, \ldots, u_m) \in \{\pm 1\}^m \tag{2.6}$$

associated with the edges. A gauge transformations $D_l \in \{D_1, \ldots, D_n\}$ maps between such tuples by flipping the variables on the edges incident to site l. We call two configurations equivalent if they are related to each other by a gauge transformation. This induces an equivalence relation and a partition of the configuration space into disjoint equivalence classes $[\boldsymbol{u}]$.

Proposition 1. *Let $\mathcal{G}$ be a connected graph. Then, there are $m - n + 1$ non-empty equivalence classes $[\boldsymbol{u}]$ and as many independent cycles γ_l in $\mathcal{G}$, such that each $[\boldsymbol{u}]$ can*

be uniquely labeled by the gauge-invariant tuple $\left(W_1([\boldsymbol{u}]), \ldots, W_{m-n+1}([\boldsymbol{u}])\right)$ *with*

$$W_l([\boldsymbol{u}]) \coloneqq \prod_{\text{edge } j \in \gamma_l} u_j \qquad \text{for any representative } \boldsymbol{u} \text{ of } [\boldsymbol{u}]. \tag{2.7}$$

Proof. To prove the 1-to-1 correspondence between W_l-labels and equivalence classes $[\boldsymbol{u}]$, the proof will be structured into two steps: First, eliminate $n-1$ degrees of freedom using the gauge freedom. Then, we show that the remaining $m-n+1$ degrees of freedom allow to realize each of the 2^{m-n+1} configurations for the $W_l([\boldsymbol{u}])$.

Step 1: Show that every class $[\boldsymbol{u}]$ has a representative of the form

$$\boldsymbol{u} = \big(u_1, \ldots, u_{m-n+1}, \underbrace{1, \ldots, 1}_{\text{n-1 times}}\big) \in \{\pm 1\}^m . \tag{2.8}$$

The following algorithm transforms any configuration $\boldsymbol{u}$ into one sufficing equation 2.8. Consider a spanning tree $\mathcal{T}$ of the graph, i.e. a subgraph $\mathcal{T} \subset \mathcal{G}$ connecting all vertices of $\mathcal{G}$ without containing a cycle. Such a tree always exists for a connected graph and contains $n-1$ edges. We pick our edge-labeling such that these correspond to $u_{m-n+2}, \ldots, u_m$. Now let $\boldsymbol{u}$ be an arbitrary configuration and apply the following steps:

1: Pick some starting vertex $v_{j=1}$ in the spanning tree, mark it as visited.

2: Pick some next unvisited vertex v_{j+1} adjacent to a visited one (e.g. following a depth-first-search). Label the variable on the traversed edge as u_{m-n+j}.

3: Potentially apply a gauge transformation D_{j+1} at the new vertex $j+1$ to assert $u_{m-n+j} = 1$ and mark v_{j+1} as visited.

5: While $j < n$: Repeat from step 2 with $j \to j+1$.

This algorithm traverses every edge of the tree $\mathcal{T}$, installs the correct value $u_{m-n+j} = 1$ and is guaranteed not to affect the same edge again, as both vertices of this edge are marked as visited. With that, $n-1$ degrees of freedom are eliminated.

Next, we show that the remaining $m-n+1$ degrees of freedom of configurations satisfying eq. 2.8 always allow to realize each possible combination of the $m-n+1$ W_l-values defined in eq. 2.7. We will show that this is true for some specifically constructed set of independent loops γ_l.

Let $(W_1, \ldots, W_{m-n+1}) \in \{\pm 1\}^{m-n+1}$ be fixed. We describe a way to pick the values $u_1, \ldots, u_{m-n+1}$ to obtain a configuration $\boldsymbol{u}$ with these W_l-values. For conciseness, we

construct the cycles γ_l during this procedure, but note that these are independent of the desired W_l-values. Add the remaining edges $u_1, \ldots, u_{m-n+1}$ to the spanning tree $\mathcal{T}$ inductively to recover the original graph $\mathcal{G}$ and keep track of all added cycles. Start with an empty set of cycles $\Gamma = \{\}$ and $l = 0$.

Induction step $l - 1 \to l$: Pick an edge u_l of $\mathcal{G}$ that is currently missing in $\mathcal{T}$. So far, only the edges $u_1, \ldots, u_{l-1}$ have been added to $\mathcal{T}$, so none of the cycles in Γ can contain u_l. Now, add u_l to $\mathcal{T}$. With that, new cycles containing u_l become possible. Let γ_l be one of them, and add it to Γ. As γ_l is the only cycle in Γ containing the edge u_l, the value $u_l = \pm 1$ can be picked such that $W_{\gamma_l} = 1$ without affecting the already fixed $W_1, \ldots, W_{l-1}$.

After the $m - n + 1$ induction steps needed to recover $\mathcal{G}$ from $\mathcal{T}$, we have constructed $m - n + 1$ independent cycles in Γ and a state $\boldsymbol{u}$ obeying eq. 2.8 and possessing the desired W_l-values around these cycles. To sum up, out of the m degrees of freedom of a configuration $\boldsymbol{u}$, $n - 1$ can be eliminated using the gauge freedom. The remaining 2^{m-n+1} configurations realize the 2^{m-n+1} different combinations of W_l-values, and are therefore in 1-to-1 correspondence with these. The W_l are gauge-invariant, since in closed loops, each vertex must be included in an even number of edges, therefore canceling the sign-flips of any gauge transformation. This concludes the prove that there are exactly 2^{m-n+1} gauge-inequivalent classes of configurations.

□

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